How-to-CNU-notebooks

Every week you'll work through a Jupyter notebook, e.g. `CNU_Week3.ipynb`. You run them online on **JupyterHub.nrw**, so you don't need to install anything on your own computer. The images and data come from the course **Sciebo** folder; the notebooks come from the course **GitHub** repository.

Before the first session

Make sure you have access to the course Sciebo folder.

You should have received a share invitation for the folder `CNU`. Log in to Sciebo in your browser and check that `CNU` shows up in your files. If it's listed as a pending share, accept it first. If you don't see it at all, please let me know.

1. Start your JupyterHub server

1. Go to https://www.jupyterhub.nrw/en/ and choose **University of Münster** (or go straight to https://uni-muenster.jupyterhub.nrw).
2. Log in with your university account.
3. On the start page, select these options:
    - **CPU:** 1.0 – 8.0 vCPU
    - **Memory:** 4 GiB – 8 GiB
    - **Image:** Data Science + Machine Learning
        - Python, R, CUDA, PyTorch, TensorFlow → **latest version**
4. **Connect your Sciebo account** on the same page and allow access when Sciebo asks.
5. Start the server. The first start can take a minute or two.

When JupyterLab opens, you should see a folder called `sciebo` in the file browser on the left. Inside it is `CNU`, with one subfolder per week (`Week2`, `Week3`, …).

2. Get the notebooks

In JupyterLab, open **File** → **New** → **Terminal** and run:

```
git clone https://github.com/ekin-doga/Cognitive_Neuroscience_Update.git
```

(Type or paste this as **one line**.)

This creates a folder `Cognitive_Neuroscience_Update` with one folder per week, e.g. `Week3-Lindsay2021/CNU_Week3.ipynb`.

You only do this **once**. Don't run `git clone` again. It would fail, because the folder already exists.

Before each session, get the latest version by running this in a terminal:

```
cd ~/Cognitive_Neuroscience_Update
git restore .
git pull
```

This downloads new weeks and any changes to earlier notebooks. The notebooks may be updated during the semester (fixes, new exercises), so pull every time.

`git restore .` resets the notebooks to their original state. Running a notebook saves its outputs into the file, and without this step `git pull` would refuse to update. This also removes your outputs and any edits you made to the notebooks.

`git pull` only updates the notebooks.

Note: If you want to keep a copy of the notebook with your outputs, download it before running these commands.

3. Run the first notebook

1. Go here in the folder tree: /Cognitive_Neuroscience_Update/Week2-Intro-AlexNet/
2. Click on CNU_Week2.ipynb
3. Click into the first cell and press the run button ▶ (marked with a red rectangle), or press **Shift + Enter**. This runs the cell and moves to the next one. Go through the notebook cell by cell and check that each cell runs without an error.

